

T3 Lagrangian and Weinberg-operator report

EFT order: 5; loop order: 1.

$$T3 - d3 - d5 - F4, \quad \alpha = 0 : \quad S_1 = (3, 0), \quad S_2 = (5, 1), \quad F = (4, \frac{1}{2})$$

Build T3 ingredients	Success Success	Matching Weinberg extraction	Success Success
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Accepted interactions

Yukawa1_GenericCG, Yukawa2_GenericCG, Scalar1SelfInv1, Scalar1SelfInv2, Scalar2SelfInv1, Scalar2SelfInv2, Scalar2SelfInv3, HiggsPortal1Inv1, HiggsPortal1Inv2, HiggsPortal2Inv1, HiggsPortal2Inv2, ScalarCrossPortalInv1, T3ScalarMix_GenericCG.

BSM free Lagrangian $\mathcal{L}_{\text{free}}$

$$\begin{aligned} & i\overline{N_{j_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{j_1, r_1}) \\ & - M_F \overline{N_{j_1, r_1}} N_{j_1, r_1} \\ & + D_{\mu_1} (\text{NewScalar1}_{j_1}) D_{\mu_1} (\text{NewScalar1}_{j_1})^\dagger \\ & + D_{\mu_1} (\text{NewScalar2}_{j_1}) D_{\mu_1} (\text{NewScalar2}_{j_1})^\dagger \\ & + M_{S1}^2 (-\text{NewScalar1}_{j_1}) \text{NewScalar1}_{j_1}^\dagger \\ & - M_{S2}^2 \text{NewScalar2}_{j_1} \text{NewScalar2}_{j_1}^\dagger \end{aligned}$$

Interaction Lagrangian $\mathcal{L}_{\text{int}}$

$$\begin{aligned} & \overline{\ell_{a_1, i_1}} C P_R \overline{N_{j_1, r_1}}^T \text{NewScalar1}_{j_1} \text{T3Y1CG}_{a_1, j_1, j_1} y_{1, i_1, r_1} \\ & + \overline{y_{2, i_1, r_1}} \overline{N_{j_1, r_1}} P_L \ell_{a_1, i_1} \text{NewScalar2}_{j_1} \overline{\text{T3Y2CG}}_{a_1, j_1, j_1} \\ & + \overline{y_{1, i_1, r_1}} \ell_{a_1, i_1}^T C P_L N_{j_1, r_1} \overline{\text{T3Y1CG}}_{a_1, j_1, j_1} \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH1Inv1} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv1}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaH1Inv2} H_{a_2} \text{NewScalar1}_{j_2} \text{T3HiggsPortal1CGInv2}_{a_1, a_2, j_2, j_1} H_{a_1}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \overline{\lambda_{\text{T3}}} \text{NewScalar2}_{j_1} \overline{\text{T3MixCG}}_{a_1, a_2, j_1, j_1} H_{a_1}^\dagger H_{a_2}^\dagger \text{NewScalar1}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaS1Inv1} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\ & + \frac{1}{2} \text{lambdaS1Inv2} \text{NewScalar1}_{j_3} \text{NewScalar1}_{j_4} \text{T3Scalar1SelfCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar1}_{j_2}^\dagger \\ & + \overline{\ell_{a_1, i_1}} P_R N_{j_1, r_1} \text{T3Y2CG}_{a_1, j_1, j_1} y_{2, i_1, r_1} \text{NewScalar2}_{j_1}^\dagger \\ & + H_{a_1} H_{a_2} \text{NewScalar1}_{j_1} \text{T3MixCG}_{a_1, a_2, j_1, j_1} \lambda_{\text{T3}} \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambdaH2Inv1} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv1}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambdaH2Inv2} H_{a_2} \text{NewScalar2}_{j_2} \text{T3HiggsPortal2CGInv2}_{a_1, a_2, j_1, j_2} H_{a_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \text{lambda12Inv1} \text{NewScalar1}_{j_2} \text{NewScalar2}_{j_2} \text{T3CrossScalarCGInv1}_{j_1, j_2, j_1, j_2} \text{NewScalar1}_{j_1}^\dagger \text{NewScalar2}_{j_1}^\dagger \\ & + \frac{1}{2} \text{lambdaS2Inv1} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv1}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \frac{1}{2} \text{lambdaS2Inv2} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv2}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \\ & + \frac{1}{2} \text{lambdaS2Inv3} \text{NewScalar2}_{j_3} \text{NewScalar2}_{j_4} \text{T3Scalar2SelfCGInv3}_{j_1, j_3, j_2, j_4} \text{NewScalar2}_{j_1}^\dagger \text{NewScalar2}_{j_2}^\dagger \end{aligned}$$

BSM-induced EFT contribution $\Delta\mathcal{L}_{\text{EFT}}$

$$\begin{aligned}
& - \frac{\sqrt{\frac{3}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{2(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^4}{2(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^4}{2(M_F^2 - M_{S_1}^2)^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^4}{2(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{2\sqrt{\frac{5}{3}} \hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} \ell_{a_4,i_2}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{5}{3}} \hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{y_{1,i_1,r_1}} \overline{y_{2,i_2,r_1}} \ell_{a_4,i_2}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F^3}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\sqrt{\frac{5}{3}} \hbar \log\left(\frac{M_F^2}{M_{S_2}^2}\right) \overline{\lambda_{T3}} \ell_{a_3,i_1} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{5}{3}} \hbar \log\left(\frac{M_F^2}{M_{S_1}^2}\right) \overline{\lambda_{T3}} \ell_{a_3,i_1} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F^3}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{2(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1}} \overline{\ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_2}^2 y_{2,i_2,r_1} Y_{e,i_3,i_1} M_F^2}{(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S_1}^2} \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& + \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_1}^2 y_{1,i_3,r_1} H_{a_1}^\dagger M_F^2}{(M_F^2 - M_{S_1}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{M_F^2 - M_{S_2}^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{2(M_F^2 - M_{S_2}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L \ell_{a_1,i_1} M_{S_2}^2 y_{2,i_3,r_1} H_{a_1}^\dagger M_F^2}{(M_F^2 - M_{S_2}^2)^2} \\
& - \frac{2\sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} \ell_{a_4,i_2}}^T C P_L \ell_{a_2,i_1} H_{a_1} H_{a_3} M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \bar{e}_{a_1,a_2} \bar{e}_{a_3,a_4} M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{2\sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S_1}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& + \frac{2\sqrt{\frac{5}{3}} \hbar \log \left(\frac{M_F^2}{M_{S_2}^2} \right) \overline{\lambda_{T3} \ell_{a_3,i_1}} C P_R \overline{\ell_{a_4,i_2}}^T M_{S_1}^2 M_{S_2}^2 y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2 - M_{S_1}^2)(M_F^2 - M_{S_2}^2)(M_{S_1}^2 - M_{S_2}^2)} \\
& - \frac{17\hbar B_{\mu_1\mu_2}^2}{12\epsilon} \\
& - \hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) B_{\mu_1\mu_2}^2 \\
& - \frac{5}{12} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_2}^2} \right) B_{\mu_1\mu_2}^2 \\
& - \frac{6\hbar (W_{\mu_1\mu_2}^{A_1})^2}{\epsilon} \\
& - 5\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& - \frac{1}{6} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_1}^2} \right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& - \frac{5}{6} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_2}^2} \right) (W_{\mu_1\mu_2}^{A_1})^2 \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2\epsilon} \\
& - \frac{1}{4} \sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1} \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{M_F^2 - M_{S_1}^2} \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_1}^2} \right) \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^2 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{M_F^2 - M_{S_1}^2} \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i_3,r_1} \ell_{a_1,i_2}} P_R e_{i_1} H_{a_1} M_{S_1}^4 y_{1,i_2,r_1} Y_{e,i_3,i_1}}{2(M_F^2 - M_{S_1}^2)^2}
\end{aligned}$$

$$\begin{aligned}
& - \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S1}^2} \right) \overline{y_{1,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{S1}^4 y_{1,i2,r1} Y_{e,i3,i1}}{2 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} y_{2,i2,r1} Y_{e,i3,i1}}{2\epsilon} \\
& - \frac{1}{4} \sqrt{\frac{5}{2}} \hbar \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} y_{2,i2,r1} Y_{e,i3,i1} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{S2}^2 y_{2,i2,r1} Y_{e,i3,i1}}{M_F^2 - M_{S2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S2}^2} \right) \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{S2}^2 y_{2,i2,r1} Y_{e,i3,i1}}{M_F^2 - M_{S2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{S2}^4 y_{2,i2,r1} Y_{e,i3,i1}}{2 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S2}^2} \right) \overline{y_{2,i3,r1} \ell_{a1,i2}} P_R e_{i1} H_{a1} M_{S2}^4 y_{2,i2,r1} Y_{e,i3,i1}}{2 (M_F^2 - M_{S2}^2)^2} \\
& + C_{H2} H_{a1} H_{a1}^\dagger \\
& + \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} y_{1,i3,r1} H_{a1}^\dagger}{2\epsilon} \\
& - \frac{1}{4} \sqrt{\frac{3}{2}} \hbar \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} y_{1,i3,r1} H_{a1}^\dagger \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{S1}^2 y_{1,i3,r1} H_{a1}^\dagger}{M_F^2 - M_{S1}^2} \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S1}^2} \right) \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{S1}^2 y_{1,i3,r1} H_{a1}^\dagger}{M_F^2 - M_{S1}^2} \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{S1}^4 y_{1,i3,r1} H_{a1}^\dagger}{2 (M_F^2 - M_{S1}^2)^2} \\
& - \frac{\sqrt{\frac{3}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S1}^2} \right) \overline{y_{1,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{S1}^4 y_{1,i3,r1} H_{a1}^\dagger}{2 (M_F^2 - M_{S1}^2)^2} \\
& + \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} y_{2,i3,r1} H_{a1}^\dagger}{2\epsilon} \\
& - \frac{1}{4} \sqrt{\frac{5}{2}} \hbar \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} y_{2,i3,r1} H_{a1}^\dagger \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{S2}^2 y_{2,i3,r1} H_{a1}^\dagger}{M_F^2 - M_{S2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S2}^2} \right) \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{S2}^2 y_{2,i3,r1} H_{a1}^\dagger}{M_F^2 - M_{S2}^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{S2}^4 y_{2,i3,r1} H_{a1}^\dagger}{2 (M_F^2 - M_{S2}^2)^2} \\
& - \frac{\sqrt{\frac{5}{2}} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S2}^2} \right) \overline{y_{2,i1,r1} Y_{e,i3,i2} \bar{e}_{i2}} P_L \ell_{a1,i1} M_{S2}^4 y_{2,i3,r1} H_{a1}^\dagger}{2 (M_F^2 - M_{S2}^2)^2} \\
& - H_{a1} \mu^2 H_{a1}^\dagger \\
& + \frac{3 \text{lambdaH1Inv} 2^2 \hbar H_{a1} H_{a2} H_{a1}^\dagger H_{a2}^\dagger}{2\epsilon}
\end{aligned}$$

$$\begin{aligned}
& + \frac{15 \text{lambdaH2Inv1}^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{14\epsilon} \\
& + \frac{95 \text{lambdaH2Inv2}^2 \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{42\epsilon} \\
& + \frac{10 \sqrt{\frac{2}{3}} \text{lambdaH2Inv1} \text{lambdaH2Inv2} \hbar H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{7\epsilon} \\
& + \frac{3}{2} \text{lambdaH1Inv2}^2 \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S1}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{15}{14} \text{lambdaH2Inv1}^2 \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{95}{42} \text{lambdaH2Inv2}^2 \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{10}{7} \sqrt{\frac{2}{3}} \text{lambdaH2Inv1} \text{lambdaH2Inv2} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& + \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S1}^2} \right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S1}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2} \\
& - \frac{2 \sqrt{\frac{5}{3}} \hbar \log \left(\frac{\mu \text{bar} 2}{M_{S2}^2} \right) \overline{\lambda_{T3}} H_{a_1} H_{a_2} M_{S2}^2 \lambda_{T3} H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2 - M_{S2}^2}
\end{aligned}$$

Weinberg sector

$$\frac{2 \sqrt{\frac{5}{3}} \hbar M_F \left(M_F^2 \left(M_{S1}^2 \log \left(\frac{M_F^2}{M_{S1}^2} \right) - M_{S2}^2 \log \left(\frac{M_F^2}{M_{S2}^2} \right) \right) + M_{S1}^2 M_{S2}^2 \left(\log \left(\frac{M_F^2}{M_{S1}^2} \right) - \log \left(\frac{M_F^2}{M_{S2}^2} \right) \right) \right) \left(\lambda_{T3} H_{a1} H_{a3} \overline{\epsilon_{a1,a2}} \overline{y_{1,i1,r1}} \overline{\epsilon_{a3,a4}} \overline{y_{2,i2,r1}} \overline{\ell_{a4,i2}} \right)^T C P_L \ell_{a2,i1} + \overline{\lambda_{T3}} \epsilon_{a1,a3} y_{1,i1,r1} \epsilon_{a2,a4} H_{a1}^\dagger H_{a2}^\dagger y_{2,i2,r1} \overline{\ell_{a3,i1}} C P_R \overline{\ell_{a4,i2}}^T}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)}$$

Weinberg coefficient C_5

$$\frac{2}{3} \left(- \frac{\sqrt{15} \hbar M_F^3 M_{S2}^2 \lambda_{T3} \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y_{1,i1,r1}} \overline{y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{15} \hbar M_F^3 M_{S1}^2 \lambda_{T3} \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y_{1,i1,r1}} \overline{y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} - \frac{\sqrt{15} \hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log \left(\frac{M_F^2}{M_{S1}^2} \right) \overline{y_{1,i1,r1}} \overline{y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{15} \hbar M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log \left(\frac{M_F^2}{M_{S2}^2} \right) \overline{y_{1,i1,r1}} \overline{y_{2,i2,r1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \right)$$